Reflection Analysis

Input Parameters

Generel

Data file:	*16-3-3 SWL Berm 4% - test 2.experiment - 60% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.469 Hz*
Freq. upper bound:	*9.375 Hz*

Freq. Domain Analysis

% overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*19*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*20*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *16.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.06919*	*0.01172*
Peak wave period T_P (s):	*1.067*	*1.123*
Energy wave period $T_{-1,0}$ (s):	*1.055*	*0.7938*
Mean wave period $T_{0,1}$ (s):	*1.023*	*0.6669*
Mean wave period $T_{0,2}$ (s):	*1.003*	*0.6161*
Spectral width (Broadness):	*0.4718*	*0.657*
Spectral width (Narrowness):	*0.1986*	*0.4143*

Time Domain Analysis

Overall amplitude reflection coefficient: *17%*

	Incident	**Reflected**
Number of waves:	1005	1499
Mean wave height H_m (m):	*0.04288*	*0.007153*
Mean wave period T_m (s):	*1.003*	*0.6721*
Sig. wave height H_s (m):	*0.06844*	*0.01163*
$T_{H_{1/3}}$ (s):	*1.046*	*0.7631*
H_{max} (m):	*0.1302*	*0.0291*
T_{Hmax} (s):	*0.989*	*0.5575*
$H_{1/10}$ (m):	*0.0847*	*0.01523*
$H_{1/50}$ (m):	*0.1007*	*0.01997*
$H_{1/100}$ (m)	*0.1069*	*0.02245*
$H_{1/250}$ (m):	*0.1165*	*0.02547*
$H_{0.1\%}$ (m):	*0.1302*	*0.0278*
$H_{1\%}$ (m):	*0.09824*	*0.01912*
$H_{2\%}$ (m):	*0.09249*	*0.01651*
$H_{10\%}$ (m)	*0.0739*	*0.01237*
Groupiness factor GF:	*0.9629*	*1.248*
Skewness b1:	*0.03219*	*0.03733*
Kurtosis b2:	*2.926*	*3.803*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*11. mrt 2026*
Remarks	
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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 11.0% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*24*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *24.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1396*	*0.03449*
Peak wave period T_P (s):	*1.524*	*0.6531*
Energy wave period $T_{-1,0}$ (s):	*1.402*	*0.9554*
Mean wave period $T_{0,1}$ (s):	*1.298*	*0.7221*
Mean wave period $T_{0,2}$ (s):	*1.222*	*0.6487*
Spectral width (Broadness):	*0.7667*	*0.7323*
Spectral width (Narrowness):	*0.3575*	*0.4893*

Time Domain Analysis

Overall amplitude reflection coefficient: *24.7%*

	Incident	**Reflected**
Number of waves:	939	1541
Mean wave height H_m (m):	*0.09147*	*0.02262*
Mean wave period T_m (s):	*1.329*	*0.8094*
Sig. wave height H_s (m):	*0.137*	*0.03478*
$T_{H_{1/3}}$ (s):	*1.46*	*0.8879*
H_{max} (m):	*0.1916*	*0.07209*
T_{Hmax} (s):	*1.442*	*1.548*
$H_{1/10}$ (m):	*0.1607*	*0.04395*
$H_{1/50}$ (m):	*0.1796*	*0.05413*
$H_{1/100}$ (m)	*0.184*	*0.05826*
$H_{1/250}$ (m):	*0.1899*	*0.06348*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06918*
$H_{1\%}$ (m):	*0.1774*	*0.05275*
$H_{2\%}$ (m):	*0.1724*	*0.04838*
$H_{10\%}$ (m)	*0.1455*	*0.03702*
Groupiness factor GF:	*0.882*	*1.166*
Skewness b1:	*0.4675*	*0.1398*
Kurtosis b2:	*2.749*	*3.558*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 120% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.320 Hz*
Freq. upper bound:	*6.406 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*30*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *25.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1462*	*0.03738*
Peak wave period T_P (s):	*1.561*	*1.829*
Energy wave period $T_{-1,0}$ (s):	*1.451*	*1.034*
Mean wave period $T_{0,1}$ (s):	*1.336*	*0.7562*
Mean wave period $T_{0,2}$ (s):	*1.252*	*0.6716*
Spectral width (Broadness):	*0.7788*	*0.7417*
Spectral width (Narrowness):	*0.3723*	*0.5178*

Time Domain Analysis

Overall amplitude reflection coefficient: *25.4%*

	Incident	**Reflected**
Number of waves:	945	1554
Mean wave height H_m (m):	*0.09735*	*0.02453*
Mean wave period T_m (s):	*1.388*	*0.8438*
Sig. wave height H_s (m):	*0.1442*	*0.03754*
$T_{H_{1/3}}$ (s):	*1.527*	*0.9509*
H_{max} (m):	*0.2055*	*0.06281*
T_{Hmax} (s):	*1.913*	*0.6023*
$H_{1/10}$ (m):	*0.1681*	*0.04655*
$H_{1/50}$ (m):	*0.1904*	*0.05508*
$H_{1/100}$ (m)	*0.1979*	*0.05747*
$H_{1/250}$ (m):	*0.2022*	*0.06009*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06181*
$H_{1\%}$ (m):	*0.1905*	*0.05479*
$H_{2\%}$ (m):	*0.1794*	*0.05117*
$H_{10\%}$ (m)	*0.1526*	*0.0407*
Groupiness factor GF:	*0.8777*	*1.132*
Skewness b1:	*0.5088*	*0.1775*
Kurtosis b2:	*2.727*	*3.499*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 1300 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.313 Hz*
Freq. upper bound:	*6.250 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*31*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *27%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1576*	*0.04252*
Peak wave period T_P (s):	*1.6*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.508*	*1.123*
Mean wave period $T_{0,1}$ (s):	*1.376*	*0.7936*
Mean wave period $T_{0,2}$ (s):	*1.279*	*0.693*
Spectral width (Broadness):	*0.8029*	*0.7673*
Spectral width (Narrowness):	*0.395*	*0.5579*

Time Domain Analysis

Overall amplitude reflection coefficient: *27%*

	Incident	**Reflected**
Number of waves:	949	1547
Mean wave height H_m (m):	*0.1044*	*0.02774*
Mean wave period T_m (s):	*1.441*	*0.8838*
Sig. wave height H_s (m):	*0.1518*	*0.04179*
$T_{H_{1/3}}$ (s):	*1.583*	*1.015*
H_{max} (m):	*0.2251*	*0.07963*
T_{Hmax} (s):	*1.672*	*1.81*
$H_{1/10}$ (m):	*0.177*	*0.05233*
$H_{1/50}$ (m):	*0.2008*	*0.0639*
$H_{1/100}$ (m)	*0.2088*	*0.06853*
$H_{1/250}$ (m):	*0.2182*	*0.07378*
$H_{0.1\%}$ (m):	*Not enough data*	*0.07906*
$H_{1\%}$ (m):	*0.1974*	*0.06194*
$H_{2\%}$ (m):	*0.1873*	*0.05758*
$H_{10\%}$ (m)	*0.1605*	*0.04414*
Groupiness factor GF:	*0.8538*	*1.098*
Skewness b1:	*0.5596*	*0.1591*
Kurtosis b2:	*2.679*	*3.325*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 140% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *29.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1646*	*0.04802*
Peak wave period T_P (s):	*1.73*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.554*	*1.228*
Mean wave period $T_{0,1}$ (s):	*1.401*	*0.8399*
Mean wave period $T_{0,2}$ (s):	*1.293*	*0.7232*
Spectral width (Broadness):	*0.8117*	*0.7757*
Spectral width (Narrowness):	*0.4168*	*0.5906*

Time Domain Analysis

Overall amplitude reflection coefficient: *29.2%*

	Incident	**Reflected**
Number of waves:	955	1514
Mean wave height H_m (m):	*0.1116*	*0.03194*
Mean wave period T_m (s):	*1.495*	*0.9427*
Sig. wave height H_s (m):	*0.1588*	*0.04814*
$T_{H_{1/3}}$ (s):	*1.626*	*1.15*
H_{max} (m):	*0.2169*	*0.0908*
T_{Hmax} (s):	*1.666*	*1.377*
$H_{1/10}$ (m):	*0.1834*	*0.05962*
$H_{1/50}$ (m):	*0.2027*	*0.07229*
$H_{1/100}$ (m)	*0.2086*	*0.07729*
$H_{1/250}$ (m):	*0.2124*	*0.08187*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08975*
$H_{1\%}$ (m):	*0.2023*	*0.07042*
$H_{2\%}$ (m):	*0.1935*	*0.06474*
$H_{10\%}$ (m)	*0.1668*	*0.05097*
Groupiness factor GF:	*0.8223*	*1.045*
Skewness b1:	*0.5683*	*0.1486*
Kurtosis b2:	*2.597*	*3.185*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 150 Wave.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1712*	*0.05296*
Peak wave period T_P (s):	*1.73*	*2.133*
Energy wave period $T_{-1,0}$ (s):	*1.596*	*1.307*
Mean wave period $T_{0,1}$ (s):	*1.423*	*0.8775*
Mean wave period $T_{0,2}$ (s):	*1.305*	*0.7457*
Spectral width (Broadness):	*0.8232*	*0.7942*
Spectral width (Narrowness):	*0.4361*	*0.6202*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.9%*

	Incident	**Reflected**
Number of waves:	920	1511
Mean wave height H_m (m):	*0.1186*	*0.03453*
Mean wave period T_m (s):	*1.551*	*0.9446*
Sig. wave height H_s (m):	*0.164*	*0.05182*
$T_{H_{1/3}}$ (s):	*1.703*	*1.187*
H_{max} (m):	*0.2208*	*0.09229*
T_{Hmax} (s):	*1.537*	*2.02*
$H_{1/10}$ (m):	*0.187*	*0.06493*
$H_{1/50}$ (m):	*0.2055*	*0.07791*
$H_{1/100}$ (m)	*0.2118*	*0.08286*
$H_{1/250}$ (m):	*0.218*	*0.08817*
$H_{0.1\%}$ (m):	*Not enough data*	*0.09202*
$H_{1\%}$ (m):	*0.2052*	*0.07634*
$H_{2\%}$ (m):	*0.1963*	*0.07038*
$H_{10\%}$ (m)	*0.1717*	*0.05586*
Groupiness factor GF:	*0.809*	*1.039*
Skewness b1:	*0.6156*	*0.174*
Kurtosis b2:	*2.585*	*3.192*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3 SWL Berm 4% - test 2.experiment - 80% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.406 Hz*
Freq. upper bound:	*8.125 Hz*

Freq. Domain Analysis

% of length of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*21*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*24*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *22.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1061*	*0.02351*
Peak wave period T_P (s):	*1.231*	*0.6531*
Energy wave period $T_{-1,0}$ (s):	*1.213*	*0.8194*
Mean wave period $T_{0,1}$ (s):	*1.158*	*0.672*
Mean wave period $T_{0,2}$ (s):	*1.119*	*0.6195*
Spectral width (Broadness):	*0.6598*	*0.6925*
Spectral width (Narrowness):	*0.2677*	*0.4205*

Time Domain Analysis

Overall amplitude reflection coefficient: *22.2%*

	Incident	**Reflected**
Number of waves:	980	1596
Mean wave height H_m (m):	*0.0669*	*0.01452*
Mean wave period T_m (s):	*1.151*	*0.7066*
Sig. wave height H_s (m):	*0.1031*	*0.0236*
$T_{H_{1/3}}$ (s):	*1.228*	*0.7856*
H_{max} (m):	*0.1471*	*0.05463*
T_{Hmax} (s):	*1.064*	*0.6071*
$H_{1/10}$ (m):	*0.1226*	*0.03073*
$H_{1/50}$ (m):	*0.1374*	*0.0403*
$H_{1/100}$ (m)	*0.1418*	*0.04408*
$H_{1/250}$ (m):	*0.1452*	*0.04916*
$H_{0.1\%}$ (m):	*Not enough data*	*0.053*
$H_{1\%}$ (m):	*0.1378*	*0.03797*
$H_{2\%}$ (m):	*0.1312*	*0.03496*
$H_{10\%}$ (m)	*0.1102*	*0.025*
Groupiness factor GF:	*0.8928*	*1.337*
Skewness b1:	*0.1542*	*0.3208*
Kurtosis b2:	*2.778*	*4.17*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-3-SWL Berm 4% - test 2.experiment - 100% (4) (x)*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.344 Hz*
Freq. upper bound:	*6.875 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*23*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*28*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *23.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1299*	*0.03091*
Peak wave period T_P (s):	*1.455*	*0.6957*
Energy wave period $T_{-1,0}$ (s):	*1.345*	*0.9168*
Mean wave period $T_{0,1}$ (s):	*1.258*	*0.7098*
Mean wave period $T_{0,2}$ (s):	*1.195*	*0.6419*
Spectral width (Broadness):	*0.7382*	*0.7246*
Spectral width (Narrowness):	*0.3289*	*0.4721*

Time Domain Analysis

Overall amplitude reflection coefficient: *23.9%*

	Incident	**Reflected**
Number of waves:	935	1546
Mean wave height H_m (m):	*0.08349*	*0.01983*
Mean wave period T_m (s):	*1.27*	*0.7682*
Sig. wave height H_s (m):	*0.1262*	*0.03063*
$T_{H_{1/3}}$ (s):	*1.388*	*0.8455*
H_{max} (m):	*0.1843*	*0.06809*
T_{Hmax} (s):	*1.538*	*0.4981*
$H_{1/10}$ (m):	*0.1482*	*0.03928*
$H_{1/50}$ (m):	*0.1683*	*0.0499*
$H_{1/100}$ (m)	*0.174*	*0.05482*
$H_{1/250}$ (m):	*0.1795*	*0.06114*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06597*
$H_{1\%}$ (m):	*0.1671*	*0.0479*
$H_{2\%}$ (m):	*0.1594*	*0.0426*
$H_{10\%}$ (m)	*0.1327*	*0.03241*
Groupiness factor GF:	*0.8942*	*1.197*
Skewness b1:	*0.3957*	*0.1744*
Kurtosis b2:	*2.755*	*3.651*

Department of Civil Engineering

Aalborg University

Project title	
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Remarks	
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